WESTERN GRAY WOLF CASE FILE

Shared evidence packet - all stakeholder groups receive the same core evidence

THE DECISION

Gray wolves have recovered substantially across portions of the western United States. Assume the U.S. Fish and Wildlife Service (FWS) is considering whether federal Endangered Species Act (ESA) protections should be removed from a western gray-wolf population.

Should FWS delist it?

Your task: Use the shared evidence below, plus your stakeholder-specific case file, to support or oppose delisting. Distinguish evidence relevant to ESA status from other social, economic, or political concerns.

1. Population recovery

Gray wolves were essentially eliminated from the western United States by the 1930s. Recovery came from natural recolonization from Canada and reintroductions into central Idaho and Yellowstone National Park in 1995-1996. Wolves then expanded through the Northern Rockies and into Oregon, Washington, California, and Colorado. By the end of 2022, FWS estimated **2,797 wolves in more than 286 packs across seven western states**. Long-term abundance and occupied range have increased strongly since reintroduction, although Idaho and Montana showed modest decreases in 2021-2022. [1]

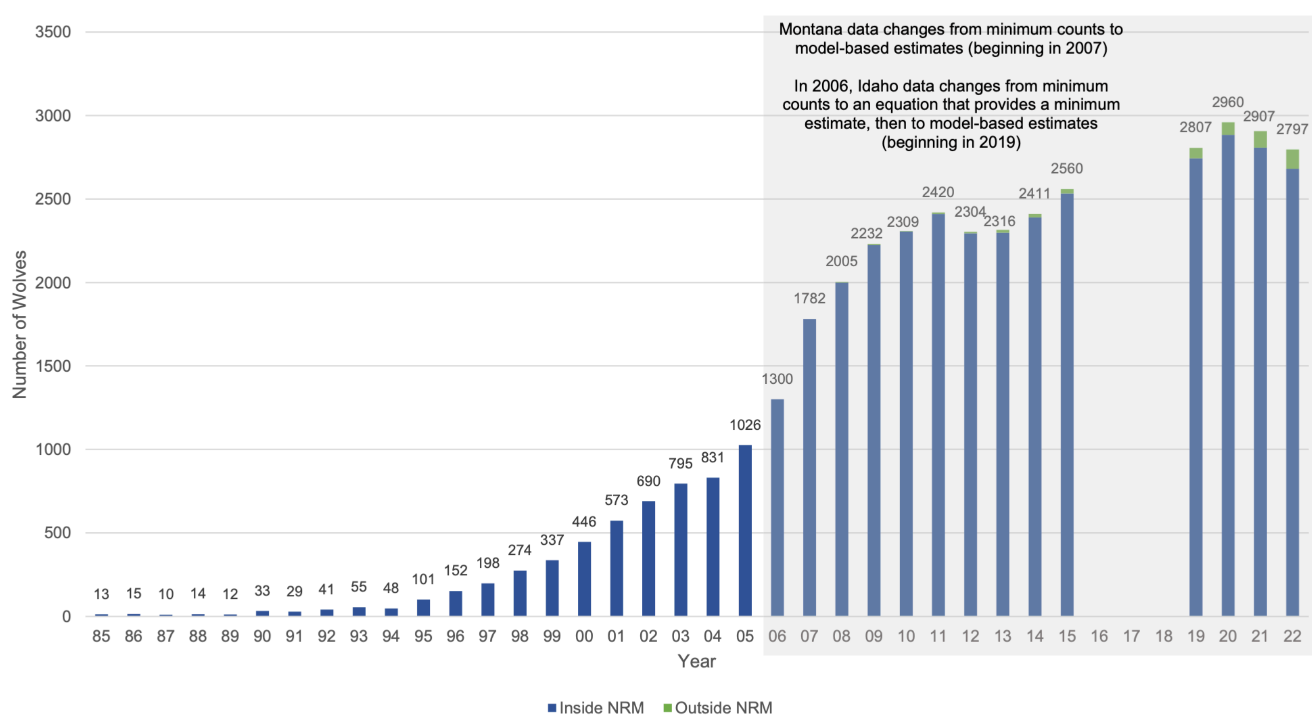

Figure 1: Sourced from Gray Wolf SSA, Figure 9. Minimum number of gray wolves counted or estimated in the Western United States, 1985–2022, both inside of the Northern Rocky Mountain (NRM) population (blue) and outside of the NRM (green). Total number of wolves in the Western United States metapopulation indicated at the top of each year's bar. These estimates do not include Mexican wolves. Note that this graph does not include estimates for the total number of wolves in 2016, 2017, or 2018, as we do not have population estimates from Idaho for these years so could not produce a total estimate for the Western United States

2. Recovery benchmarks

The Northern Rocky Mountain recovery program ultimately used a benchmark of at least **30 breeding pairs and 300 wolves** distributed among Idaho, Montana, and Wyoming for three consecutive years, together with genetic exchange among recovery populations. Western wolves have substantially exceeded the numerical criteria. However, meeting a recovery-plan threshold does *not* automatically trigger delisting: FWS must still determine whether the entity meets the ESA definitions of endangered or threatened using current evidence and the statutory threat factors. [1,3]

3. Historical and current range

Wolves historically occupied most of the western United States. Present distribution remains concentrated in the Northern Rockies and Pacific Northwest, with newer populations farther south and west. FWS treats current range as central to present status while also considering loss of historical range when evaluating threats and conservation. The meaning of "significant portion of its range" has repeatedly been contested in gray-wolf litigation. [1,3]

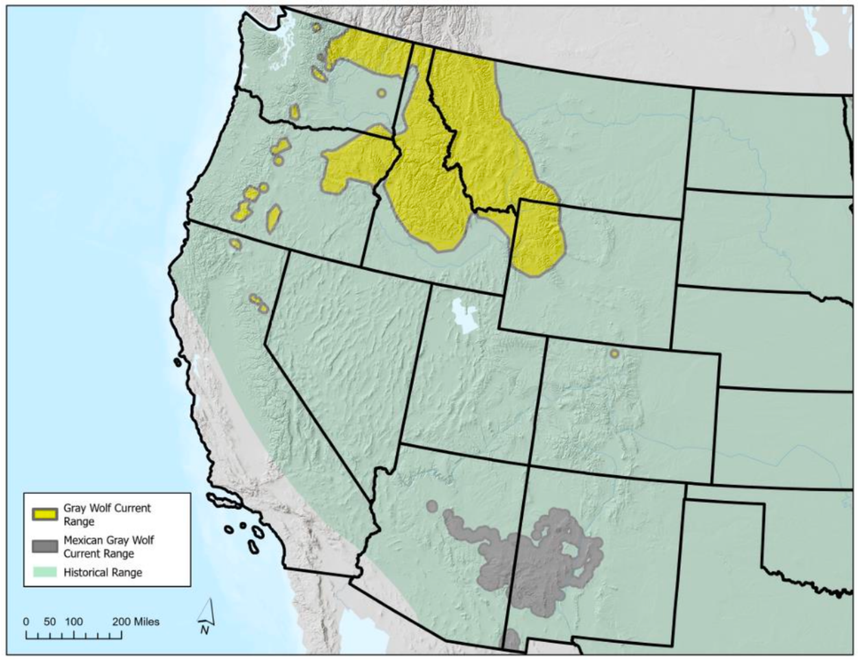

Figure 2: Sources from the Gray Wolf SSA, Figure 2. Historical (green) and current (yellow) gray wolf range in the Western United States. The U.S. portion of Mexican wolf range is depicted in gray.

4. Population structure and connectivity

The biological geography of western wolves has changed as recovery progressed. In 2024 FWS concluded that the old Northern Rocky Mountains (NRM) distinct population segment (DPS) is **no longer discrete** from wolves farther west: Oregon, Washington, and California wolves largely descend from NRM/Canadian founders, suitable habitat links these areas, and long-distance dispersal occurs. FWS instead concluded that a broader **Western United States population could qualify as a valid DPS** because it is discrete from other major wolf groups and significant to the species' range. [2]

5. Genetic health

FWS concluded that western wolves currently have high genetic diversity and connectivity, supporting resiliency and adaptive capacity. Its modeling found very low probability of decline to levels associated with short-term inbreeding risk under the modeled scenarios. [1] A genomic study by vonHoldt et al., however, found declining heterozygosity through time in Northern Rocky Mountain wolves and estimated effective population size at only about **5.2-9.3% of census size**. The authors concluded that wolves were above short-term inbreeding thresholds but below commonly proposed sizes for minimizing long-term evolutionary risk. [4]

6. Current and foreseeable threats

FWS identifies **human-caused mortality as the primary stressor** on western wolves, including regulated hunting/trapping, lethal control after livestock depredation, and illegal take. Disease can cause episodic localized declines. FWS considers habitat and prey broadly available and did not identify climate change or habitat modification as primary current threats to western wolf viability. [1]

7. What changes after delisting?

Delisting shifts primary management authority from the federal ESA framework to states and Tribes. State approaches differ substantially. Idaho and Montana expanded hunting and trapping opportunities beginning in 2021 while retaining management mechanisms and commitments intended to keep populations above federal recovery thresholds. The adequacy of state regulatory mechanisms has repeatedly been central to gray-wolf delisting decisions and litigation. [1-3]

Does the evidence show that western gray wolves are secure without federal ESA protection, or do uncertainty about distribution, genetics, mortality, and future management justify continued protection?

Sources

[1] U.S. Fish and Wildlife Service. 2023. Species Status Assessment for the Gray Wolf (Canis lupus) in the Western United States, Version 1.2. Especially Executive Summar and Chapters 2-4.

[2] U.S. Fish and Wildlife Service. 2024. Species Assessment and Listing Priority Assignment Form / 12-month finding for gray wolf in the Northern Rocky Mountains and Western United States (information current Jan. 4, 2024).

[3] Congressional Research Service. 2020. The Gray Wolf Under the Endangered Species Act (ESA): A Case Study in Listing and Delisting Challenges. R46184.

[4] vonHoldt, B.M. et al. 2024. Demographic history shapes North American gray wolf genomic diversity and informs species' conservation. Molecular Ecology 33:e17231. doi:10.1111/mec.17231.

[5] Martin, D.R.P. 2020. Prolonged gray wolf endangered species act listing fluctuation and policymaker response in the United States. Biological Conservation 242:108408.

[6] U.S. Fish and Wildlife Service. 2024. 'Service Announces Gray Wolf Finding and National Recovery Plan.' Feb. 2, 2024.